

Credit Name: CSE 2140 2nd Language Programming  
Assignment Name: Chapter 4 Mastery Exercise 6 (MathTutor)

**The following questions can help you in thinking critically about your problem-solving processes:**

#### **Understanding the Problem**

How did you approach understanding the challenge?

The IPO chart was the main thing that helped me understand that 2 numbers and an operator had to be randomly generated as an equation first, then displayed, then accept the users answer and then determine what the correct answer was, and if the user was right. The final output tells the user if they're correct or incorrect.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

It took me a while to figure the best way to randomly generate an equation, but the biggest problem was getting the division equations to match the difficulty of the other possible equations. I had to ask AI how to make it so the division always resulted in an integer.

#### **Planning the Solution**

Did you create a plan or break the problem into smaller steps before coding?

I tried to break it down into smaller, testable parts. Such as generating 2 random numbers from 1-10 and creating a random operator to use with the 2 numbers.

How did you decide on the tools, data structures, or algorithms to use?

I used integer variables because the numbers could only be from 1 -10. I used the Random class to randomly generate those numbers and to also help randomize the operator. If else statements helped create random operators and then what the correct answer would be depending on the operator.

#### **Implementation**

Did you write the code in small pieces or attempt the entire solution at once?

This code was longer so I wrote it in small peices to make sure I was on the right track.

How did you test your solution along the way to make sure it was working?

I would run a few lines of test code to see if it worked. This would have helped me catch any errors early and be able to fix them without entirely rewriting my code.

#### **Overcoming Challenges**

What part of the problem was the most difficult for you?

Generating the random operator and making sure that the division equations resulted in an integer

How did you handle moments when you felt stuck or unsure of what to do next?

I used the resources around me such as the textbook, friends and online tools.

#### **Learning**

Was there anything you learned that you think will help you with future challenges?

In this code I got the hang of randomly generating numbers or even a string, practiced with if else statements and discovered an application of the while loop. This practice should help me considerably in the future.